

7

Independence

7.1 Independent events and random variables

Although a probability space is nothing more than a measure space with the measure of the whole space equal to one, probability theory is not merely a subset of measure theory. A distinguishing and fundamental feature of probability theory is the notion of *independence*.

Definition 7.1.1: Let $(\Omega, \mathcal{F}, P)$ be a probability space and $\{B_1, B_2, \dots, B_n\} \subset \mathcal{F}$ be a finite collection of events.

(i) $B_1, B_2, \dots, B_n$ are called *independent* w.r.t. P , if

$$P\left(\bigcap_{j=1}^k B_{i_j}\right) = \prod_{j=1}^k P(B_{i_j}) \quad (1.1)$$

for all $\{i_1, i_2, \dots, i_k\} \subset \{1, 2, \dots, n\}, 1 \leq k \leq n$.

(ii) $B_1, B_2, \dots, B_n$ are called *pairwise independent* w.r.t. P if $P(B_i \cap B_j) = P(B_i)P(B_j)$ for all $i, j, i \neq j$.

Note that a collection $B_1, B_2, \dots, B_n$ of events may be independent with respect to one probability measure P but not with respect to another measure P' . Note also that pairwise independence does not imply independence (Problem 7.1).

Definition 7.1.2: Let $(\Omega, \mathcal{F}, P)$ be a probability space. A collection of events $\{B_\alpha, \alpha \in A\} \subset \mathcal{F}$ is called *independent* w.r.t. P if for every finite

subcollection $\{\alpha_1, \alpha_2, \dots, \alpha_k\} \subset A, 1 \leq k < \infty$,

$$P\left(\bigcap_{i=1}^k B_{\alpha_i}\right) = \prod_{i=1}^k P(B_{\alpha_i}). \quad (1.2)$$

Definition 7.1.3: Let $(\Omega, \mathcal{F}, P)$ be a probability space. Let A be a nonempty set. For each α in A , let $\mathcal{G}_\alpha \subset \mathcal{F}$ be a collection of events. Then the family $\{\mathcal{G}_\alpha : \alpha \in A\}$ is called *independent* w.r.t P if for every choice of B_α in $\mathcal{G}_\alpha$ for α in A , the collection of events $\{B_\alpha : \alpha \in A\}$ is independent w.r.t. P as in Definition 7.1.2.

Definition 7.1.4: Let $(\Omega, \mathcal{F}, P)$ be a probability space and let $\{X_\alpha : \alpha \in A\}$ be a collection of random variables on $(\Omega, \mathcal{F}, P)$. Then the collection $\{X_\alpha : \alpha \in A\}$ is called *independent* w.r.t. P if the family of σ -algebras $\{\sigma\langle X_\alpha \rangle : \alpha \in A\}$ is independent w.r.t. P , where $\sigma\langle X_\alpha \rangle$ is the σ -algebra generated by X_α , i.e.,

$$\sigma\langle X_\alpha \rangle \equiv \{X_\alpha^{-1}(B) : B \in \mathcal{B}(\mathbb{R})\}. \quad (1.3)$$

Note that the collection $\{X_\alpha : \alpha \in A\}$ is independent iff for any $\{\alpha_1, \alpha_2, \dots, \alpha_k\} \subset A$, and $B_i \in \mathcal{B}(\mathbb{R})$, for $i = 1, 2, \dots, k$, $1 \leq k < \infty$,

$$P(X_{\alpha_i} \in B_i, i = 1, 2, \dots, k) = \prod_{i=1}^k P(X_{\alpha_i} \in B_i). \quad (1.4)$$

It turns out that if (1.4) holds for all B_i of the form $B_i = (-\infty, x_i]$, $x_i \in \mathbb{R}$, then it holds for all $B_i \in \mathcal{B}(\mathbb{R})$, $i = 1, 2, \dots, k$. This follows from the proposition below.

Proposition 7.1.1: Let $(\Omega, \mathcal{F}, P)$ be a probability space. Let A be a nonempty set. Let $\mathcal{G}_\alpha \subset \mathcal{F}$ be a π -system for each α in A . Let $\{\mathcal{G}_\alpha : \alpha \in A\}$ be independent w.r.t. P . Then the family of σ -algebras $\{\sigma\langle \mathcal{G}_\alpha \rangle : \alpha \in A\}$ is also independent w.r.t. P .

Proof: Fix $2 \leq k < \infty$, $\{\alpha_1, \alpha_2, \dots, \alpha_k\} \subset A$, $B_i \in \mathcal{G}_{\alpha_i}$, $i = 1, 2, \dots, k-1$. Let

$$\mathcal{L} \equiv \left\{ B : B \in \sigma\langle \mathcal{G}_{\alpha_k} \rangle, P(B_1 \cap \dots \cap B_{k-1} \cap B) = \left(\prod_{i=1}^{k-1} P(B_i) \right) P(B) \right\}. \quad (1.5)$$

It is easy to verify that $\mathcal{L}$ is a λ -system. By hypothesis, $\mathcal{L}$ contains the π -system $\mathcal{G}_{\alpha_k}$. Hence by the π - λ theorem (cf. Theorem 1.1.2), $\mathcal{L} = \sigma\langle \mathcal{G}_{\alpha} \rangle$. Iterating the above argument k times completes the proof. $\square$

Corollary 7.1.2: *A collection $\{X_{\alpha} : \alpha \in A\}$ of random variables on a probability space $(\Omega, \mathcal{F}, P)$ is independent w.r.t. P iff for any $\{\alpha_1, \alpha_2, \dots, \alpha_k\} \subset A$ and any $x_1, x_2, \dots, x_k$ in $\mathbb{R}$, the joint cdf $F_{\alpha_1, \alpha_2, \dots, \alpha_k}$ of $(X_{\alpha_1}, X_{\alpha_2}, \dots, X_{\alpha_k})$ is the product of the marginal cdfs F_{α_i} , i.e.,*

$$\begin{aligned} F_{\alpha_1, \alpha_2, \dots, \alpha_k}(x_1, x_2, \dots, x_k) &\equiv P(X_{\alpha_i} \leq x_i, i = 1, 2, \dots, k) \\ &= \prod_{i=1}^k P(X_{\alpha_i} \leq x_i) = \prod_{i=1}^k F_{\alpha_i}(x_i). \end{aligned} \quad (1.6)$$

Proof: For the ‘if’ part let $\mathcal{G}_{\alpha} \equiv \{X_{\alpha}^{-1}((-\infty, x]) : x \in \mathbb{R}\}$, $\alpha \in A$. Now apply Proposition 7.1.1. The only if part is easy. $\square$

Remark 7.1.1: If the probability distribution of $(X_{\alpha_1}, X_{\alpha_2}, \dots, X_{\alpha_k})$ is absolutely continuous w.r.t. the Lebesgue measure m_k on $\mathbb{R}^k$, then (1.6) and hence the independence of $\{X_{\alpha_1}, X_{\alpha_2}, \dots, X_{\alpha_k}\}$ is equivalent to the condition that

$$f_{\alpha_1, \alpha_2, \dots, \alpha_k}(x_1, x_2, \dots, x_k) = \prod_{i=1}^k f_{\alpha_i}(x_i), \quad (1.7)$$

a.e. (m_k) , where $f_{(\alpha_1, \alpha_2, \dots, \alpha_k)}$ is the joint density of $(X_{\alpha_1}, X_{\alpha_2}, \dots, X_{\alpha_k})$, and f_{α_i} is the marginal density of X_{α_i} , $i = 1, 2, \dots, k$. See Problem 7.18.

Proposition 7.1.3: *Let $(\Omega, \mathcal{F}, P)$ be a probability space and let $\{X_1, X_2, \dots, X_k\}$, $2 \leq k < \infty$ be a collection of random variables on $(\Omega, \mathcal{F}, P)$.*

(i) *Then $\{X_1, X_2, \dots, X_k\}$ is independent iff*

$$E \prod_{i=1}^k h_i(X_i) = \prod_{i=1}^k E h_i(X_i) \quad (1.8)$$

for all bounded Borel measurable functions $h_i : \mathbb{R} \rightarrow \mathbb{R}$, $i = 1, 2, \dots, k$.

(ii) *If X_1, X_2 are independent and $E|X_1| < \infty, E|X_2| < \infty$, then*

$$E|X_1 X_2| < \infty \quad \text{and} \quad E X_1 X_2 = E X_1 E X_2. \quad (1.9)$$

Proof:

(i) If (1.8) holds, then taking $h_i = I_{B_i}$ with $B_i \in \mathcal{B}(\mathbb{R})$, $i = 1, 2, \dots, k$ yields the independence of $\{X_1, X_2, \dots, X_k\}$. Conversely,

if $\{X_1, X_2, \dots, X_k\}$ are independent, then (1.8) holds for $h_i = I_{B_i}$ for $B_i \in \mathcal{B}(\mathbb{R})$, $i = 1, 2, \dots, k$, and hence for simple functions $\{h_1, h_2, \dots, h_k\}$. Now (1.8) follows from the BCT.

(ii) Note that by the change of variable formula (Proposition 6.2.1)

$$\begin{aligned} E|X_1 X_2| &= \int_{\mathbb{R}^2} |x_1 x_2| dP_{X_1, X_2}(x_1, x_2), \\ E|X_i| &= \int_{\mathbb{R}} |x_i| dP_{X_i}(x_i), \quad i = 1, 2, \end{aligned}$$

where P_{X_1, X_2} is the joint distribution of (X_1, X_2) and P_{X_i} is the marginal distribution of X_i , $i = 1, 2$. Also, by the independence of X_1 and X_2 , P_{X_1, X_2} is equal to the product measure $P_{X_1} \times P_{X_2}$. Hence, by Tonelli's theorem,

$$\begin{aligned} E|X_1 X_2| &= \int_{\mathbb{R}^2} |x_1 x_2| dP_{X_1, X_2}(x_1, x_2) \\ &= \int_{\mathbb{R}^2} |x_1 x_2| dP_{X_1}(x_1) dP_{X_2}(x_2) \\ &= \left(\int_{\mathbb{R}} |x_1| dP_{X_1}(x_1) \right) \left(\int_{\mathbb{R}} |x_2| dP_{X_2}(x_2) \right) \\ &= E|X_1| E|X_2| < \infty. \end{aligned}$$

Now using Fubini's theorem, one gets (1.9). $\square$

Remark 7.1.2: Note that the converse to (ii) above need not hold. That is, if X_1 and X_2 are two random variables such that $E|X_1| < \infty$, $E|X_2| < \infty$, $E|X_1 X_2| < \infty$, and $EX_1 X_2 = EX_1 EX_2$, then X_1 and X_2 need not be independent.

7.2 Borel-Cantelli lemmas, tail σ -algebras, and Kolmogorov's zero-one law

In this section some basic results on classes of independent events are established. These will play an important role in proving laws of large numbers in Chapter 8.

Definition 7.2.1: Let $(\Omega, \mathcal{F})$ be a measurable space and $\{A_n\}_{n \geq 1}$ be a sequence of sets in $\mathcal{F}$. Then

$$\limsup_{n \rightarrow \infty} A_n \equiv \overline{\lim} A_n \equiv \bigcap_{k=1}^{\infty} \left(\bigcup_{n \geq k} A_n \right) \quad (2.1)$$

$$\liminf_{n \rightarrow \infty} A_n \equiv \underline{\lim} A_n \equiv \bigcup_{k=1}^{\infty} \bigcap_{n \geq k} A_n. \quad (2.2)$$

Proposition 7.2.1: Both $\overline{\lim} A_n$ and $\underline{\lim} A_n \in \mathcal{F}$ and

$$\begin{aligned} \overline{\lim} A_n &= \{\omega : \omega \in A_n \text{ for infinitely many } n\} \\ \underline{\lim} A_n &= \{\omega : \omega \in A_n \text{ for all but a finite number of } n\}. \end{aligned}$$

Proof: Since $\{A_n\}_{n \geq 1} \subset \mathcal{F}$ and $\mathcal{F}$ is a σ -algebra, $B_k = \bigcup_{n \geq k} A_n \in \mathcal{F}$ for each $k \in \mathbb{N}$ and hence $\overline{\lim} A_n \equiv \bigcap_{k=1}^{\infty} B_k \in \mathcal{F}$. Next,

$$\begin{aligned} \omega \in \overline{\lim} A_n &\iff \omega \in B_k \text{ for all } k = 1, 2, \dots \\ &\iff \text{for each } k, \text{ there exists } n_k \geq k \text{ such that } \omega \in A_{n_k} \\ &\iff \omega \in A_n \text{ for infinitely many } n. \end{aligned}$$

The proof for $\underline{\lim} A_n$ is similar. $\square$

In probability theory, $\overline{\lim} A_n$ is referred to as the event that “ A_n happens infinitely often (i.o.)” and $\underline{\lim} A_n$ as the event that “all but a finitely many A_n ’s happen.”

Example 7.2.1: Let $\Omega = \mathbb{R}$, $\mathcal{F} = \mathcal{B}(\mathbb{R})$, and let

$$A_n = \begin{cases} [0, \frac{1}{n}] & \text{for } n \text{ odd} \\ [1 - \frac{1}{n}, 1] & \text{for } n \text{ even.} \end{cases}$$

Then $\overline{\lim} A_n = \{0, 1\}$, $\underline{\lim} A_n = \emptyset$.

The following result on the probabilities of $\overline{\lim} A_n$ and $\underline{\lim} A_n$ is very useful in probability theory.

Theorem 7.2.2: Let $(\Omega, \mathcal{F}, P)$ be a probability space and $\{A_n\}_{n \geq 1}$ be a sequence of events in $\mathcal{F}$. Then

(a) (The first Borel-Cantelli lemma). If $\sum_{n=1}^{\infty} P(A_n) < \infty$, then $P(\overline{\lim} A_n) = 0$.

(b) (The second Borel-Cantelli lemma). If $\sum_{n=1}^{\infty} P(A_n) = \infty$ and $\{A_n\}_{n \geq 1}$ are pairwise independent, then $P(\overline{\lim} A_n) = 1$.

Remark 7.2.1: This result is also called a *zero-one law* as it asserts that for pairwise independent events $\{A_n\}_{n \geq 1}$, $P(\overline{\lim} A_n) = 0$ or 1 according to $\sum_{n=1}^{\infty} P(A_n) < \infty$ or equal to ∞ .

Proof:

- (a) Let $Z_n \equiv \sum_{j=1}^n I_{A_j}$. Then $Z_n \uparrow Z \equiv \sum_{j=1}^{\infty} I_{A_j}$ and by the MCT, $EZ_n \equiv \sum_{j=1}^n P(A_j) \uparrow EZ$. Thus, $\sum_{j=1}^{\infty} P(A_j) < \infty \Rightarrow EZ < \infty \Rightarrow Z < \infty$ w.p. 1 $\Rightarrow P(Z = \infty) = 0$. But the event $\overline{\lim} A_n = \{Z = \infty\}$ and so (a) follows.
- (b) Without loss of generality, assume $P(A_j) > 0$ for some j . Let $J_n = \frac{Z_n}{EZ_n}$ for $n \geq j$ where Z_n is as above. Then, $EJ_n = 1$ and by the pairwise independence of $\{A_n\}_{n \geq 1}$, the variance of J_n is

$$\text{Var}(J_n) = \frac{\sum_{j=1}^n P(A_j)(1 - P(A_j))}{(EZ_n)^2} \leq \frac{1}{(EZ_n)}.$$

If $\sum_{j=1}^{\infty} P(A_j) = \infty$, then $EZ_n = \sum_{j=1}^n P(A_j) \uparrow \infty$, by the MCT. Thus $EJ_n \equiv 1$, $\text{Var}(J_n) \rightarrow 0$ as $n \rightarrow \infty$. By Chebychev's inequality, for all $\epsilon > 0$,

$$P(|J_n - 1| > \epsilon) \leq \frac{\text{Var}(J_n)}{\epsilon^2} \rightarrow 0 \quad \text{as } n \rightarrow \infty.$$

Thus, $J_n \rightarrow 1$ in probability and hence there exists a subsequence $\{n_k\}_{k \geq 1}$ such that $J_{n_k} \rightarrow 1$ w.p. 1 (cf. Theorem 2.5.2). Since $EZ_{n_k} \uparrow \infty$, this implies that $Z_{n_k} \rightarrow \infty$ w.p. 1. But $\{Z_n\}_{n \geq 1}$ is nondecreasing in n and hence $Z_n \uparrow \infty$ w.p. 1. Now since $\overline{\lim} A_n = \{Z = \infty\}$, it follows that $P(\overline{\lim} A_n) = P(Z = \infty) = 1$. $\square$

Proposition 7.2.3: Let $\{X_n\}_{n \geq 1}$, be a sequence of random variables on some probability space $(\Omega, \mathcal{F}, P)$.

- (a) If $\sum_{n=1}^{\infty} P(|X_n| > \epsilon) < \infty$ for each $\epsilon > 0$, then

$$P\left(\lim_{n \rightarrow \infty} X_n = 0\right) = 1.$$

- (b) If $\{X_n\}_{n \geq 1}$ are pairwise independent and $P\left(\lim_{n \rightarrow \infty} X_n = 0\right) = 1$, then $\sum_{n=1}^{\infty} P(|X_n| > \epsilon) < \infty$ for each $\epsilon > 0$.

Proof:

- (a) Fix $\epsilon > 0$. Let $A_n = \{|X_n| > \epsilon\}$, $n \geq 1$. Then $\sum_{n=1}^{\infty} P(A_n) < \infty \Rightarrow P(\overline{\lim} A_n) = 0$, by the first Borel-Cantelli lemma (Theorem 7.2.2 (a)). But

$$\begin{aligned} (\overline{\lim} A_n)^c &= \{\omega : \text{there exists } n(\omega) < \infty \text{ such that for all} \\ &\quad n \geq n(\omega), \omega \notin A_n\} \\ &= \{\omega : \text{there exists } n(\omega) < \infty \text{ such that } |X_n(\omega)| \leq \epsilon \\ &\quad \text{for all } n \geq n(\omega)\} \\ &= B_\epsilon, \text{ say.} \end{aligned}$$

Thus, $\sum_{n=1}^{\infty} P(A_n) < \infty \Rightarrow P(B_\epsilon) = 1$. Let $B = \bigcap_{r=1}^{\infty} B_{\frac{1}{r}}$. Now note that

$$\left\{ \omega : \lim_{n \rightarrow \infty} |X_n(\omega)| = 0 \right\} = \bigcap_{r=1}^{\infty} B_{\frac{1}{r}}.$$

Since $P(B^c) \leq \sum_{r=1}^{\infty} P(B_{\frac{1}{r}}^c) = 0$, $P(B) = 1$.

- (b) Let $\{X_n\}_{n \geq 1}$ be pairwise independent and $\sum_{n=1}^{\infty} P(|X_n| > \epsilon_0) = \infty$ for some $\epsilon_0 > 0$. Let $A_n = \{|X_n| > \epsilon_0\}$. Since $\{X_n\}_{n \geq 1}$ are pairwise independent, so are $\{A_n\}_{n \geq 1}$. By the second Borel-Cantelli lemma

$$P(\overline{\lim} A_n) = 1.$$

But $\omega \in \overline{\lim} A_n \Rightarrow \limsup_{n \rightarrow \infty} |X_n| \geq \epsilon_0$ and hence $P(\lim_{n \rightarrow \infty} |X_n| = 0) = 0$. This contradicts the hypothesis that $P(\limsup_{n \rightarrow \infty} |X_n| = 0) = 1$. $\square$

Definition 7.2.2: The *tail σ -algebra* of a sequence of random variables $\{X_n\}_{n \geq 1}$ on a probability space $(\Omega, \mathcal{F}, P)$ is

$$\mathcal{T} = \bigcap_{n=1}^{\infty} \sigma(\{X_j : j \geq n\})$$

and any $A \in \mathcal{T}$ is called a *tail event*. Further, any $\mathcal{T}$ -measurable random variable is called a *tail random variable* (w.r.t. $\{X_n\}_{n \geq 1}$).

Tail events are determined by the behavior of the sequence $\{X_n\}_{n \geq 1}$ for large n and they remain unchanged if any finite subcollection of the X_n 's are dropped or replaced by another finite set of random variables. Events such as $\{\limsup_{n \rightarrow \infty} X_n < x\}$ or $\{\lim_{n \rightarrow \infty} X_n = x\}$, $x \in \mathbb{R}$, belong to $\mathcal{T}$. A remarkable result of Kolmogorov is that for any sequence of independent random variables, any tail event has probability zero or one.

Theorem 7.2.4: (*Kolmogorov's 0-1 law*). Let $\{X_n\}_{n \geq 1}$ be a sequence of independent random variables on a probability space $(\Omega, \mathcal{F}, P)$ and let $\mathcal{T}$ be the tail σ -algebra of $\{X_n\}_{n \geq 1}$. Then $P(A) = 0$ or 1 for all $A \in \mathcal{T}$.

Remark 7.2.2: Note that in Proposition 7.2.3, the event $A \equiv \{\lim_{n \rightarrow \infty} X_n = 0\}$ belongs to $\mathcal{T}$, and hence, by the above theorem, $P(A) = 0$ or 1 . Thus, proving that $P(A) \neq 1$ is equivalent to proving $P(A) = 0$. Kolmogorov's 0-1 law only restricts the possible values of tail events like A to 0 or 1, while the Borel-Cantelli lemmas (Theorem 7.2.2) provide a tool for ascertaining whether the value is either 0 or 1. On the other hand, note that Theorem 7.2.2 requires only *pairwise independence* of $\{A_n\}_{n \geq 1}$ but Kolmogorov's 0-1 law requires the *full independence* of the sequence $\{X_n\}_{n \geq 1}$.

Proof: For $n \geq 1$, define the σ -algebras $\mathcal{F}_n$ and $\mathcal{T}_n$ by $\mathcal{F}_n = \sigma(\{X_1, \dots, X_n\})$ and $\mathcal{T}_n = \sigma(\{X_{n+1}, X_{n+2}, \dots\})$. Since $X_n, n \geq 1$ are independent, $\mathcal{F}_n$ is independent of $\mathcal{T}_n$ for all $n \geq 1$. Since, for each n , $\mathcal{T} = \bigcap_{m=n}^{\infty} \mathcal{F}_m$ is a sub σ -algebra of $\mathcal{T}_n$, this implies $\mathcal{F}_n$ is independent of $\mathcal{T}$ for all $n \geq 1$ and hence $\mathcal{A} \equiv \bigcup_{n=1}^{\infty} \mathcal{F}_n$ is independent of $\mathcal{T}$. It is easy to check that $\mathcal{A}$ is an algebra (and hence, is a π -system). Hence, by Proposition 7.1.1, $\sigma(\mathcal{A})$ is independent of $\mathcal{T}$. Since $\mathcal{T}$ is also a sub- σ -algebra of $\sigma(\mathcal{A}) = \sigma(\{X_n : n \geq 1\})$, this implies $\mathcal{T}$ is independent of itself. Hence for any $B \in \mathcal{T}$,

$$P(B \cap B) = P(B) \cdot P(B),$$

which implies $P(B) = 0$ or 1 . $\square$

Definition 7.2.3: Let $(\Omega, \mathcal{F}, P)$ be a probability space and let $X : \Omega \rightarrow \bar{\mathbb{R}}$ be a $\langle \mathcal{F}, \mathcal{B}(\bar{\mathbb{R}}) \rangle$ -measurable mapping. (Recall the definition of $\mathcal{B}(\bar{\mathbb{R}})$ from (2.1.4)). Then X is called an *extended real-valued random variable* or an *$\mathbb{R}$ -valued random variable*.

Corollary 7.2.5: Let $\mathcal{T}$ be the tail σ -algebra of a sequence of independent random variables $\{X_n\}_{n \geq 1}$ on $(\Omega, \mathcal{F}, P)$ and let X be a $\langle \mathcal{T}, \mathcal{B}(\bar{\mathbb{R}}) \rangle$ -measurable $\mathbb{R}$ -valued random variable from Ω to $\mathbb{R}$. Then, there exists $c \in \mathbb{R}$ such that

$$P(X = c) = 1.$$

Proof: If $P(X \leq x) = 0$ for all $x \in \mathbb{R}$, then $P(X = +\infty) = 1$. Hence, suppose that $B \equiv \{x \in \mathbb{R} : P(X \leq x) \neq 0\} \neq \emptyset$. Since $\{X \leq x\} \in \mathcal{T}$ for all $x \in \mathbb{R}$, $P(X \leq x) = 1$ for all $x \in B$. Define $c = \inf\{x : x \in B\}$. Check that $P(X = c) = 1$. $\square$

An immediate implication of Corollary 7.2.5 is that for any sequence of *independent* random variables $\{X_n\}_{n \geq 1}$, the $\bar{\mathbb{R}}$ -valued random variables $\limsup_{n \rightarrow \infty} X_n$ and $\liminf_{n \rightarrow \infty} X_n$ are degenerate, i.e., they are constants w.p. 1.

Example 7.2.2: Let $\{X_n\}_{n \geq 1}$ be a sequence of independent random variables on $(\Omega, \mathcal{F}, P)$ with $EX_n = 0$, $EX_n^2 = 1$ for all $n \geq 1$. Let $S_n = X_1 + \dots + X_n, n \geq 1$ and $\Phi(x) = \int_{-\infty}^x (\sqrt{2\pi})^{-1} \exp(-y^2/2) dy, x \in \mathbb{R}$. If $P(S_n \leq \sqrt{n}x) \rightarrow \Phi(x)$ for all $x \in \mathbb{R}$, then

$$\limsup_{n \rightarrow \infty} \frac{S_n}{\sqrt{n}} = +\infty \quad \text{a.s.} \quad (2.3)$$

To show this, let $S = \limsup_{n \rightarrow \infty} S_n/\sqrt{n}$. First it will be shown that S is $\langle \mathcal{T}, \mathcal{B}(\bar{\mathbb{R}}) \rangle$ -measurable. For any $m \geq 1$, define the variables $T_{m,n} = (X_{m+1} + \dots + X_n)/\sqrt{n}$ and $S_{m,n} = (X_1 + \dots + X_m)/\sqrt{n}, n > m$. Note that for any fixed $m \geq 1$, $T_{m,n}$ is $\sigma(X_{m+1}, \dots)$ -measurable and $S_{m,n}(\omega) \rightarrow 0$ as

$n \rightarrow \infty$ for all $\omega \in \Omega$. Hence, for any $m \geq 1$,

$$\begin{aligned} S &= \limsup_{n \rightarrow \infty} (S_{m,n} + T_{m,n}) \\ &= \limsup_{n \rightarrow \infty} T_{m,n} \end{aligned}$$

is $\sigma\langle X_{m+1}, X_{m+2}, \dots \rangle$ -measurable. Thus, S is measurable with respect to $\mathcal{T} = \bigcap_{m=1}^{\infty} \sigma\langle X_{m+1}, X_{m+2}, \dots \rangle$. Hence, by Theorem 7.2.4, $P(S = +\infty) \in \{0, 1\}$.

If possible, now suppose that $P(S = +\infty) = 0$. Then, by Corollary 7.2.5, there exists $c \in [-\infty, \infty)$ such that $P(S = c) = 1$. Let $A_n = \{S_n > \sqrt{n}x\}$, $n \geq 1$, with $x = c + 1$. Then,

$$\begin{aligned} 0 < 1 - \Phi(x) &= \lim_{n \rightarrow \infty} P(A_n) \\ &\leq \lim_{n \rightarrow \infty} P\left(\bigcup_{m \geq n} A_m\right) \\ &= P\left(\bigcap_{n=1}^{\infty} \bigcup_{m=n}^{\infty} A_m\right) \\ &= P\left(\frac{S_n}{\sqrt{n}} > x \text{ i.o.}\right) \\ &\leq P(S \geq c + 1) = 0. \end{aligned}$$

This shows that $P(S = +\infty)$ must be 1. Also see Problem 7.16.

Remark 7.2.3: It will be shown in Chapter 11 that if $\{X_i\}_{i \geq 1}$ are independent and identically distributed (iid) random variables with $EX_1 = 0$ and $EX_1^2 = 1$, then

$$P\left(\frac{S_n}{\sqrt{n}} \leq x\right) \rightarrow \Phi(x) \quad \text{for all } x \text{ in } \mathbb{R}.$$

(This is known as the central limit theorem.) Indeed, a stronger result known as the law of the iterated logarithm holds, which says that for such $\{X_i\}_{i \geq 1}$,

$$\limsup_{n \rightarrow \infty} \frac{S_n}{\sqrt{2n \log \log n}} = +1, \quad \text{w.p. 1.}$$

7.3 Problems

7.1 Give an example of three events A_1, A_2, A_3 on some probability space such that they are pairwise independent but not independent.

(**Hint:** Consider iid random variables X_1, X_2, X_3 with $P(X_1 = 1) =$

$\frac{1}{2} = P(X_1 = 0)$ and the events $A_1 = \{X_1 = X_2\}$, $A_2 = \{X_1 = X_3\}$, $A_3 = \{X_3 = X_1\}$.)

7.2 Let $\{X_\alpha : \alpha \in A\}$ be a collection of independent random variables on some probability space $(\Omega, \mathcal{F}, P)$. For any subset $B \subset A$, let $\mathcal{X}_B \equiv \{X_\alpha : \alpha \in B\}$.

(a) Let B be a nonempty proper subset of A . Show that the collections $\mathcal{X}_B$ and $\mathcal{X}_{B^c}$ are independent, i.e., the σ -algebras $\sigma\langle\mathcal{X}_B\rangle$ and $\sigma\langle\mathcal{X}_{B^c}\rangle$ are independent w.r.t. P .

(b) Let $\{B_\gamma : \gamma \in \Gamma\}$ be a partition of A by nonempty proper subsets B_γ . Show that the family of σ -algebras $\{\sigma\langle\mathcal{X}_{B_\gamma}\rangle : \gamma \in \Gamma\}$ are independent w.r.t. P .

7.3 Let X_1, X_2 be iid standard exponential random variables, i.e.,

$$P(X_1 \in A) = \int_{A \cap (0, \infty)} e^{-x} dx, \quad A \in \mathcal{B}(\mathbb{R}).$$

Let $Y_1 = \min(X_1, X_2)$ and $Y_2 = \max(X_1, X_2) - Y_1$. Show that Y_1 and Y_2 are independent. Generalize this to the case of three iid standard exponential random variables.

7.4 Let $\Omega = (0, 1)$, $\mathcal{F} = \mathcal{B}((0, 1))$, the Borel σ -algebra on $(0, 1)$ and P be the Lebesgue measure on $(0, 1)$. For each $\omega \in (0, 1)$, let $\omega = \sum_{i=1}^{\infty} \frac{X_i(\omega)}{2^i}$ be the nonterminating binary expansion of ω .

(a) Show that $\{X_i\}_{i \geq 1}$ are iid Bernoulli ($\frac{1}{2}$) random variables, i.e., is $P(X_1 = 0) = \frac{1}{2} = P(X_1 = 1)$.

(**Hint:** Let $s_i \in \{0, 1\}$, $i = 1, 2, \dots, k$, $k \in \mathbb{N}$. Show that the set $\{\omega : 0 < \omega < 1, X_i(\omega) = s_i, 1 \leq i \leq k\}$ is an interval of length 2^{-k} .)

(b) Show that

$$Y_1 = \sum_{i=1}^{\infty} \frac{X_{2i-1}}{2^i} \quad (3.1)$$

$$Y_2 = \sum_{i=1}^{\infty} \frac{X_{2i}}{2^i} \quad (3.2)$$

are independent Uniform $(0, 1)$ random variables.

(c) Using the fact that the set $\mathbb{N} \times \mathbb{N}$ of lattice points (m, n) is in one to one correspondence with $\mathbb{N}$ itself, construct a sequence $\{Y_i\}_{i \geq 1}$ of iid Uniform $(0, 1)$ random variables such that for each j , Y_j is a function of $\{X_i\}_{i \geq 1}$.

- (d) For any cdf F , show that the random variable $X(\omega) \equiv F^{-1}(\omega)$ has cdf F , where

$$F^{-1}(u) = \inf\{x : F(x) \geq u\} \quad \text{for } 0 < u < 1. \quad (3.3)$$

- (e) Let $\{F_i\}_{i \geq 1}$ be a sequence of cdfs on $\mathbb{R}$. Using (c), construct a sequence $\{Z_i\}_{i \geq 1}$ of independent random variables on $(\Omega, \mathcal{F}, P)$ such that Z_i has cdf F_i , $i \geq 1$.
- (f) Show that the cdf of the random variable $W \equiv \sum_{i=1}^{\infty} \frac{2X_i}{3^i}$ is the Cantor function (cf. Section 4.5).
- (g) Let $p > 1$ be a positive integer. For each $\omega \in (0, 1)$ let $\omega \equiv \sum_{i=1}^{\infty} \frac{V_i(\omega)}{p^i}$ be the nonterminating p -nary expansion of ω . Show that $\{V_i\}_{i \geq 1}$ are iid and determine the distribution of V_1 .

7.5 Let $\{X_i\}_{i \geq 1}$ be a Markov chain with state space $\mathbb{S} = \{0, 1\}$ and transition probability matrix

$$Q = \begin{pmatrix} q_0 & p_0 \\ p_1 & q_1 \end{pmatrix} \quad \text{where } p_i = 1 - q_i, \quad 0 < q_i < 1, \quad i = 0, 1.$$

Let $\tau_1 = \min\{j : X_j = 0\}$ and $\tau_{k+1} = \min\{j : j > \tau_k, X_j = 0\}$, $k = 1, 2, \dots$. Note that τ_k is the time of k th visit to the state 0.

- (a) Show that $\{\tau_{k+1} - \tau_k : k \geq 1\}$ are iid random variables and independent of τ_1 .
- (b) Show that

$$P_i(\tau_1 < \infty) = 1 \quad \text{and hence} \quad P_i(\tau_k < \infty) = 1$$

for all $k \geq 2$, $i = 0, 1$ where P_i denotes the probability distribution with $X_1 = i$ w.p. 1.

(**Hint:** Show that $\sum_{k=1}^{\infty} P(\tau_1 > k \mid X_1 = i) < \infty$ for $i = 0, 1$ and use the Borel-Cantelli lemma.)

- (c) Show also that $E_i(e^{\theta_0 \tau_1}) < \infty$ for some $\theta_0 > 0$, $i = 0, 1$, where E_i denotes the expectation under P_i .

7.6 Let X_1 and X_2 be independent random variables.

- (a) Show that for any $p > 0$,

$$E|X_1 + X_2|^p < \infty \quad \text{iff} \quad E|X_1|^p < \infty, \quad E|X_2|^p < \infty.$$

Show that this is false if X_1 and X_2 are not independent.

(**Hint:** Use Fubini's theorem to conclude that $E|X_1 + X_2|^p < \infty$ implies that $E|X_1 + x_2|^p < \infty$ for some x_2 and hence $E|X_1|^p < \infty$.)

(b) Show that if $E(X_1^2 + X_2^2) < \infty$, then

$$\text{Var}(X_1 + X_2) = \text{Var}(X_1) + \text{Var}(X_2). \quad (3.4)$$

Show by an example that (3.4) need not imply the independence of X_1 and X_2 . Show also that if X_1 and X_2 take only two values each and (3.4) holds, then X_1 and X_2 are independent.

7.7 Let X_1 and X_2 be two random variables on a probability space $(\Omega, \mathcal{F}, P)$.

(a) Show that, if

$$\begin{aligned} P(X_1 \in (a_1, b_1), X_2 \in (a_2, b_2)) \\ = P(X_1 \in (a_1, b_1))P(X_2 \in (a_2, b_2)) \end{aligned} \quad (3.5)$$

for all a_1, b_1, a_2, b_2 in a dense set D in $\mathbb{R}$, then X_1 and X_2 are independent.

(**Hint:** Show that (3.5) implies that the joint cdf of (X_1, X_2) is the product of the marginal cdfs of X_1 and X_2 and use Corollary 7.1.2.)

(b) Let $f_i : \mathbb{R} \rightarrow \mathbb{R}$, $i = 1, 2$ be two one-one functions such that both f_i and f_i^{-1} are Borel measurable, $i = 1, 2$. Show that X_1 and X_2 are independent iff $f_1(X_1)$ and $f_2(X_2)$ are independent. Conclude that X_1 and X_2 are independent iff e^{X_1} and e^{X_2} are independent.

7.8 (a) Let X_1 and X_2 be two independent bounded random variables. Show that

$$E(p_1(X_1)p_2(X_2)) = (Ep_1(X_1))(Ep_2(X_2)) \quad (3.6)$$

where $p_1(\cdot)$ and $p_2(\cdot)$ are polynomials.

(b) Show that if X_1 and X_2 are bounded random variables and (3.6) holds for all polynomials $p_1(\cdot)$ and $p_2(\cdot)$, then X_1 and X_2 are independent.

(**Hint:** Use the facts that (i) continuous functions on a bounded closed interval $[a, b]$ can be approximated uniformly by polynomials, and (ii) for any interval $(c, d) \subset [a, b]$, any random variable X and $\epsilon > 0$, there exists a continuous function f on $[a, b]$ such that $E|f(X) - I_{(c,d)}(X)| < \epsilon$, provided $P(X = c \text{ or } d) = 0$.)

7.9 Let $\{X_n\}_{n \geq 1}$ be a sequence of iid random variables on a probability space $(\Omega, \mathcal{F}, P)$. Let $R = R(\omega)$ be the radius of convergence of the power series $\sum_{n=1}^{\infty} X_n r^n$.

(a) Show that R is a tail random variable.

(**Hint:** Note that

$$R = \frac{1}{\limsup_{n \rightarrow \infty} |X_n|^{1/n}} .)$$

(b) Show that if $E(\log |X_1|)^+ = \infty$, then $R = 0$ w.p. 1. and if $E(\log |X_1|)^+ < \infty$, then $R \geq 1$ w.p. 1.

(**Hint:** Apply the Borel-Cantelli lemmas to $A_n = \{|X_n| > \lambda^n\}$ for each $\lambda > 1$.)

7.10 Let $\{A_n\}_{n \geq 1}$ be a sequence of events in $(\Omega, \mathcal{F}, P)$ such that

$$\sum_{n=1}^{\infty} P(A_n \cap A_{n+1}^c) < \infty$$

and $\lim_{n \rightarrow \infty} P(A_n) = 0$. Show that

$$P(\limsup_{n \rightarrow \infty} A_n) = 0.$$

Show also that $\lim_{n \rightarrow \infty} P(A_n) = 0$ can be replaced by $\lim_{n \rightarrow \infty} P(\bigcap_{j \geq n} A_j) = 0$.

(**Hint:** Let $B_n = A_n \cap A_{n+1}^c$, $n \geq 1$, $B = \overline{\lim} B_n$, $A = \overline{\lim} A_n$. Show that $A \cap B^c \subset \underline{\lim} A_n$.)

7.11 For any nonnegative random variable X , show that $E|X| < \infty$ iff $\sum_{n=1}^{\infty} P(|X| > \epsilon n) < \infty$ for every $\epsilon > 0$.

7.12 Let $\{X_i\}_{i \geq 1}$ be a sequence of pairwise independent and identically distribution random variables.

(a) Show that $\lim_{n \rightarrow \infty} \frac{X_n}{n} = 0$ w.p. 1 iff $E|X_1| < \infty$.

(**Hint:** $E|X_1| < \infty \iff \sum_{n=1}^{\infty} P(|X_n| > \epsilon n) < \infty$ for all $\epsilon > 0$.)

(b) Show that $E(\log |X_1|)^+ < \infty$ iff

$$\left(|X_n|\right)^{1/n} \rightarrow 1 \quad \text{w.p. 1.}$$

7.13 Let $\{X_i\}_{i \geq 1}$ be a sequence of identically distributed random variables and let $M_n = \max\{|X_j| : 1 \leq j \leq n\}$.

- (a) If $E|X_1|^\alpha < \infty$ for some $\alpha \in (0, \infty)$, then show that

$$\frac{M_n}{n^{1/\alpha}} \rightarrow 0 \text{ w.p. } 1. \quad (3.7)$$

(**Hint:** Fix $\epsilon > 0$. Let $A_n = \{|X_n| > \epsilon n^{1/\alpha}\}$. Apply the first Borel-Cantelli lemma.)

- (b) Show that if $\{X_i\}_{i \geq 1}$ are iid satisfying (3.7) for some $\alpha > 0$, then $E|X_1|^\alpha < \infty$.

(**Hint:** Apply the second Borel-Cantelli lemma.)

7.14 Let X_1 and X_2 be independent random variables with distributions μ_1 and μ_2 . Let $Y = (X_1 + X_2)$.

- (a) Show that the distribution μ of Y is the convolution $\mu_1 * \mu_2$ as defined by

$$(\mu_1 * \mu_2)(A) = \int_{\mathbb{R}} \mu_1(A - x) \mu_2(dx)$$

(cf. Problem 5.12).

- (b) Show that if X_1 has a continuous distribution then so does Y .
 (c) Show that if X_1 has an absolutely continuous distribution then so does Y and that the density function of Y is given by

$$\left(\frac{d\mu}{dm}\right)(x) \equiv f_Y(x) = \int f_{X_1}(x - u) \mu_2(du)$$

where $f_{X_1}(x) \equiv \left(\frac{d\mu_1}{dm}\right)(x)$, the probability density of X_1 .

- (d) Y has a discrete distribution iff both X_1 and X_2 are discrete.

7.15 (*AR(1) series*). Let $\{X_n\}_{n \geq 0}$ be a sequence of random variables such that for some $\rho \in \mathbb{R}$,

$$X_{n+1} = \rho X_n + \epsilon_{n+1}, \quad n \geq 0$$

where $\{\epsilon_n\}_{n \geq 1}$ are independent and independent of X_0 .

- (a) Show that if $|\rho| < 1$ and $E(\log |\epsilon_1|)^+ < \infty$, then

$$\hat{X}_n \equiv \sum_{j=0}^n \rho^j \epsilon_{j+1} \quad \text{converges w.p. } 1$$

to a limit $\hat{X}_\infty$, say.

- (b) Show that under the hypothesis of (a), for any bounded continuous function $h : \mathbb{R} \rightarrow \mathbb{R}$ and for any distribution of X_0

$$Eh(X_n) \rightarrow Eh(\hat{X}_\infty).$$

(**Hint:** Show that for each $n \geq 1$, $X_n - \rho^n X_0$ and $\hat{X}_n$ have the same distribution.)

- 7.16 Establish the following generalization of Example 7.2.2. Let $\{X_n\}_{n \geq 1}$ be a sequence of independent random variables on some probability space $(\Omega, \mathcal{F}, P)$. Suppose there exists sequences $\{a_n\}_{n \geq 1}$, $\{x_n\}_{n \geq 1}$, such that $a_n \uparrow \infty$, $x_n \uparrow \infty$ and for each $k < \infty$, $\lim_n P(S_n \leq a_n x_k) \equiv F(x_k)$ exists and is < 1 . Show that $\limsup_{n \rightarrow \infty} \frac{S_n}{a_n} = +\infty$ a.s.

- 7.17 (a) Let $\{X_i\}_{i=1}^n$ be random variables on a probability space $(\Omega, \mathcal{F}, P)$ and let $P(X_1, X_2, \dots, X_n)^{-1}(\cdot)$ be dominated by the product measure $\mu \times \mu \times \dots \times \mu$ where μ is a σ -finite measure on $(\mathbb{R}, \mathcal{B}(\mathbb{R}))$ with Radon-Nikodym derivative $f(x_1, x_2, \dots, x_n)$. Show that $\{X_i\}_{i=1}^n$ are independent w.r.t. P iff $f(x_1, x_2, \dots, x_n) \equiv \prod_{i=1}^n h_i(x_i)$ for all $(x_1, x_2, \dots, x_n) \in \mathbb{R}$ where for each i , $h_i : \mathbb{R} \rightarrow \mathbb{R}$ is Borel measurable.

- (b) Use (a) to show that if (X_1, X_2) has an absolutely continuous distribution with density $f(x_1, x_2)$ then X_1 and X_2 are independent iff

$$f(x_1, x_2) = f_1(x_1)f_2(x_2)$$

where $f_i(\cdot)$ is the density of X_i .

- (c) Using (a) or otherwise conclude that if X_i , $i = 1, 2$ are both discrete random variables then X_1 and X_2 are independent iff

$$P(X_1 = a, X_2 = b) = P(X_1 = a)P(X_2 = b)$$

for all a and b .

- 7.18 Let $\{X_n\}_{n \geq 1}$ be a sequence of independent random variables such that for $n \geq 1$, $P(X_n = 1) = \frac{1}{n} = 1 - P(X_n = 0)$. Show that $X_n \xrightarrow{P} 0$ but not w.p. 1.

- 7.19 Let $(\Omega, \mathcal{F}, P)$ be a probability space.

- (a) Suppose there exists events $A_1, A_2, \dots, A_k$ that are independent with $0 < P(A_i) < 1$, $i = 1, 2, \dots, k$. Show that $|\Omega| \geq 2^k$ where for any set A , $|A|$ is the number of elements in A .
- (b) Let $\{X_i\}_{i=1}^k$ be independent random variables such that X_i takes n_i distinct values with positive probability. Show that $|\Omega| \geq \prod_{j=1}^k n_j$.

- (c) Show that there exists a probability space $(\Omega, \mathcal{F}, P)$ such that $|\Omega| = 2^k$ and k independent events $A_1, A_2, \dots, A_k$ in $\mathcal{F}$ such that $0 < P(A_i) < 1$, $i = 1, 2, \dots, k$.

- 7.20 (a) Let $\Omega \equiv \{(x_1, x_2) : x_1, x_2 \in \mathbb{R}, x_1^2 + x_2^2 \leq 1\}$ be the unit disc in $\mathbb{R}^2$. Let $\mathcal{F} \equiv \mathcal{B}(\Omega)$, the Borel σ -algebra in Ω and P = normalized Lebesgue measure, i.e., $P(A) \equiv \frac{m(A)}{\pi}$, $A \in \mathcal{F}$. For $\omega = (x_1, x_2)$ let

$$X_1(\omega) = x_1, \quad X_2(\omega) = x_2,$$

and $(R(\omega), \theta(\omega))$ be the polar representation of ω . Show that the random variables R and θ are independent but X_1 and X_2 are not.

- (b) Formulate and establish an extension of the above to the unit sphere in $\mathbb{R}^3$.

- 7.21 Let X_1, X_2, X_3 be iid random variables such that $P(X_1 = x) = 0$ for all $x \in \mathbb{R}$.

- (a) Show that for any permutation σ of $(1, 2, 3)$

$$P(X_{\sigma(1)} > X_{\sigma(2)} > X_{\sigma(3)}) = \frac{1}{3!}.$$

- (b) Show that for any $i = 1, 2, 3$

$$P(X_i = \max_{1 \leq j \leq 3} X_j) = \frac{1}{3}.$$

- (c) State and prove a generalization of (a) and (b) to random variables $\{X_i : 1 \leq i \leq n\}$ such that the joint distribution of $(X_1, X_2, \dots, X_n)$ is the same as that of $(X_{\sigma(1)}, X_{\sigma(2)}, \dots, X_{\sigma(n)})$ for any permutation σ of $\{1, 2, \dots, n\}$ and $P(X_1 = x) = 0$ for all $x \in \mathbb{R}$.

- 7.22 Let $f, g : \mathbb{R} \rightarrow \mathbb{R}$ be monotone nondecreasing. Show that for any random variable X

$$Ef(X)g(X) \geq Ef(X)Eg(X)$$

provided all the expectations exist.

(**Hint:** Let Y be independent of X with same distribution. Note that $Z = (f(X) - f(Y))(g(X) - g(Y)) \geq 0$ w.p. 1.)

- 7.23 Let $X_1, X_2, \dots, X_n$ be random variables on some probability space $(\Omega, \mathcal{F}, P)$. Show that if $P(X_1, X_2, \dots, X_n)^{-1}(\cdot) \ll m_n$, the Lebesgue measure on $\mathbb{R}^n$ then for each i , $PX_i^{-1}(\cdot) \ll m$. Give an example to show that the converse is not true. Show also that if

$P(X_1, X_2, \dots, X_n)^{-1}(\cdot) \ll m_n$ then $\{X_1, X_2, \dots, X_n\}$ are independent iff $f_{(X_1, X_2, \dots, X_n)}(x_1, x_2, \dots, x_n) = \prod_{i=1}^n f_{X_i}(x_i)$ where the f 's are the respective pdfs.